Mobile App Development Proposal: [Communi-Sight]

Prepared For: [District 1 - Barangay]

Prepared By: [Grace Angel G. Serrano, Krystine Ember M. Gonzaga]

Date: [February 10, 2026]

Version: 1.0

1. Executive Summary

In local communities, the lack of centralized, real-time communication channels leads to delayed complaint resolutions and uninformed residents. Conventional methods through physical bulletin boards, manual filing, lengthy forms, protracted phone calls, and poor follow-up suffer from inefficiency and opacity, resulting in unreported issues like potholes, broken streetlights, and littered parks that undermine governance and civic engagement.

This research addresses these challenges by introducing Communi-Sight, a digital-Android and IOS application designed to streamline community issue reporting. Developed by a team of five, Communi-Sight, enables users to quickly submit reports, view nearby issues through resident submitted address and to evaluate the issue quicker through the posted pictures of the complainants, and prioritize fixes throughout the barangay community. By replacing outdated analog systems with this innovative platform, Communi-Sight, promotes transparent, responsive local governance and empowers residents in real-time problem-solving.

2. Project Overview

2.1 The Problem

Residents frequently observe pervasive local issues such as potholes, malfunctioning streetlights, overflowing trash bins, and unsafe public spaces but underreport them due to the inconvenience of legacy systems, including cumbersome paper forms, unintuitive websites, and protracted telephone interactions with minimal status updates. Consequently, complaints are often ignored, deprioritized, or overlooked, as current mechanisms lack efficient reporting, real-time tracking, and collaborative prioritization tools.

2.2 The Solution

Communi-Sight lets users take a photo of a problem (like a pothole, broken streetlight, or trash buildup), add a short description, and the report with the submitted address in a few taps, replacing long forms and phone calls. The app then visualizes reported issues on the app so everyone can see what's happening nearby and which locations are most affected.

3. Scope of Work (SOW)

3.1 Target Platforms

The Barangay Report App will be developed for both iOS and Android platforms, ensuring accessibility to a wide range of smartphone users. The app will primarily support mobile phones, allowing residents to easily use its features on commonly used devices. This cross-platform strategy addresses the diverse device ecosystem in the Philippines, where Android holds over 80% market share but iOS is prevalent among urban professionals. By leveraging frameworks like Flutter or React Native, development efficiency is maximized while maintaining native performance for updated address, camera, and notifications.

3.2 User Roles

The app defines three primary user roles to facilitate seamless interaction between residents and local governance:

- Residents: Everyday barangay members who capture and submit issues (e.g., potholes, flooding, or stray animals) via photos, address, and track resolution status. This role empowers citizen participation without bureaucratic hurdles.
- Barangay Officials: Local captains or staff who review reports, assign tasks to teams, update statuses, and generate analytics dashboards for resource allocation.
- Captain/Response Teams: Field workers who receive assigned tickets, mark progress with photos, and confirm fixes, closing the feedback loop for transparency.

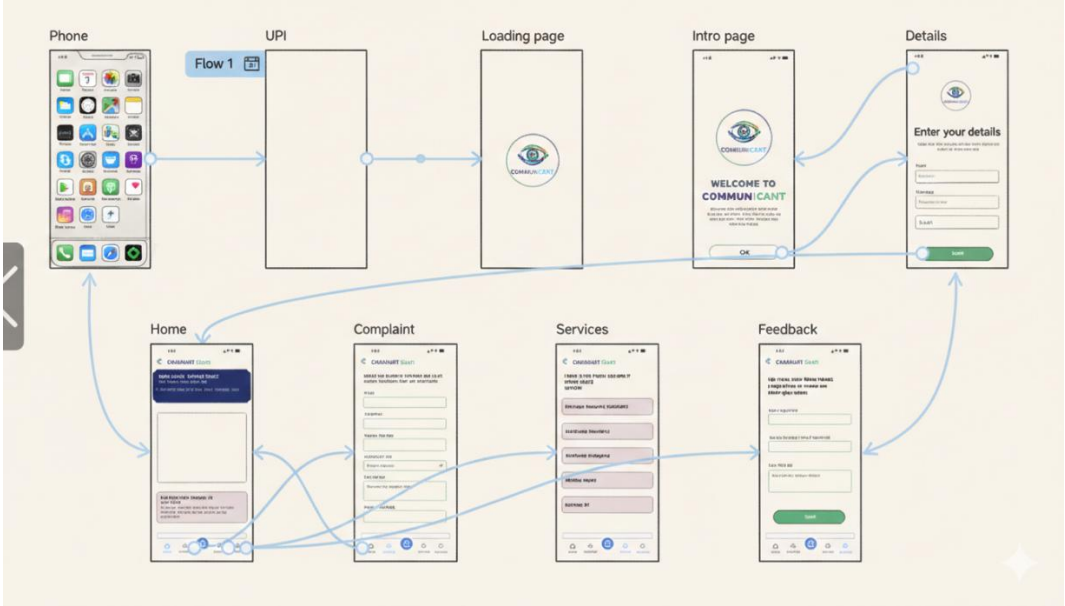
3.3 Key Features List

Communi-Sight incorporates user onboarding via secure account registration and authentication mechanisms to safeguard access. Core features enable residents to submit geo-tagged reports, including textual descriptions, photographic evidence, and precise location data. The platform integrates push notifications for real-time updates on report status and barangay announcements, thereby establishing an efficient, bidirectional communication channel between residents and officials.

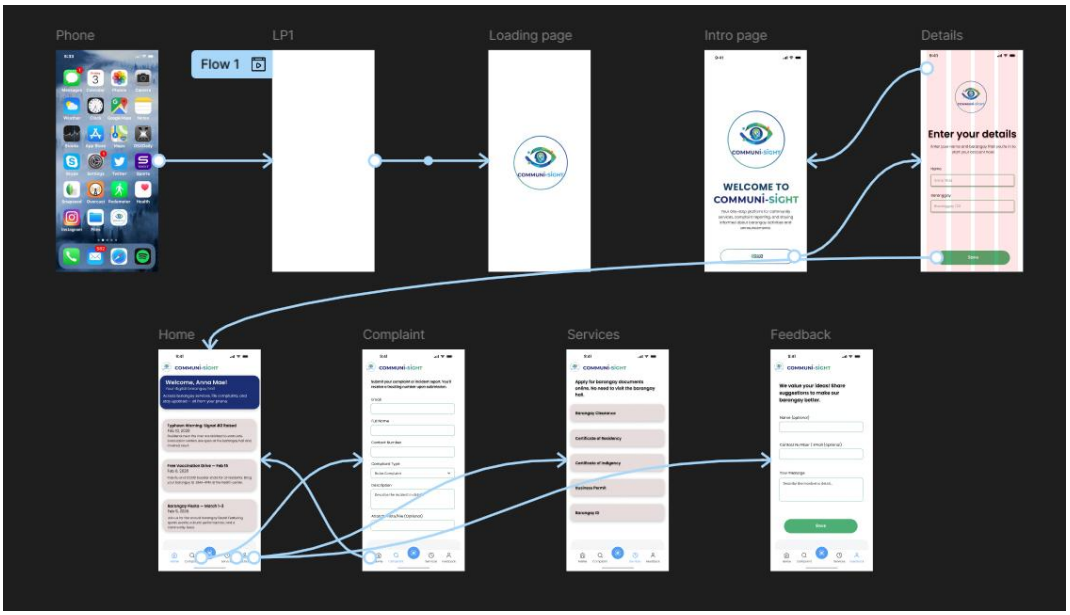
4. UI/UX Design Strategy

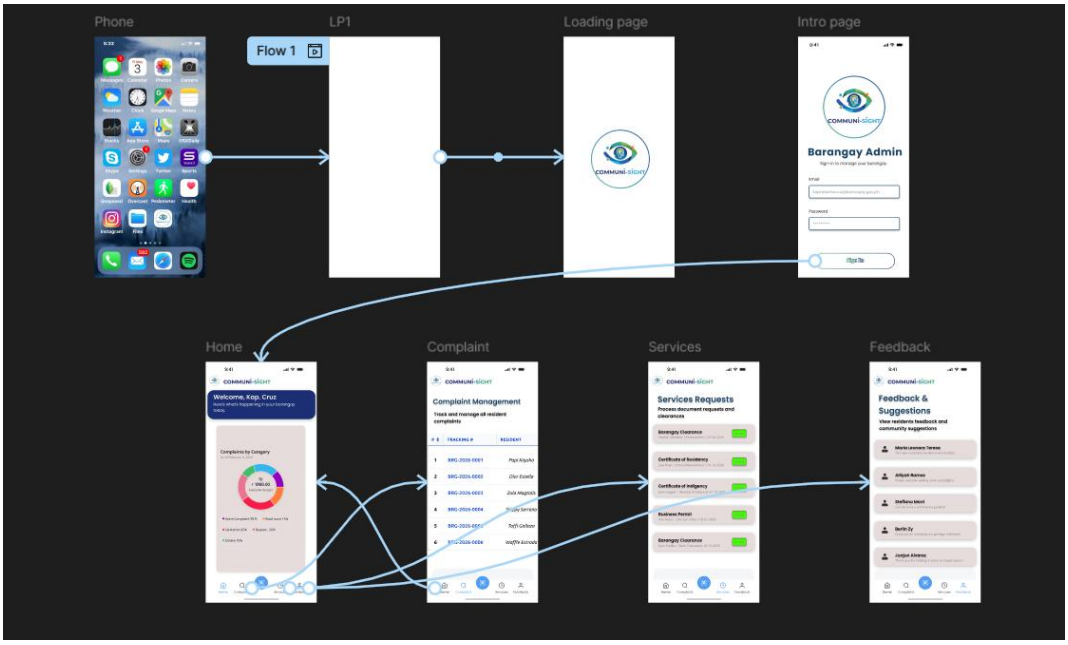
The user interface design process commences with wireframing to delineate the application's architecture and navigational flows. This is succeeded by low-fidelity prototyping via interactive mockups, enabling validation of user interactions and feature accessibility. High-fidelity designs follow, integrating barangay-specific branding elements such as color palettes and typography to yield a professional, cohesive aesthetic. Collectively, these phases aim to deliver an intuitive, accessible, and visually compelling user experience.

- **Wireframing:** Low-fidelity sketches to establish flow.

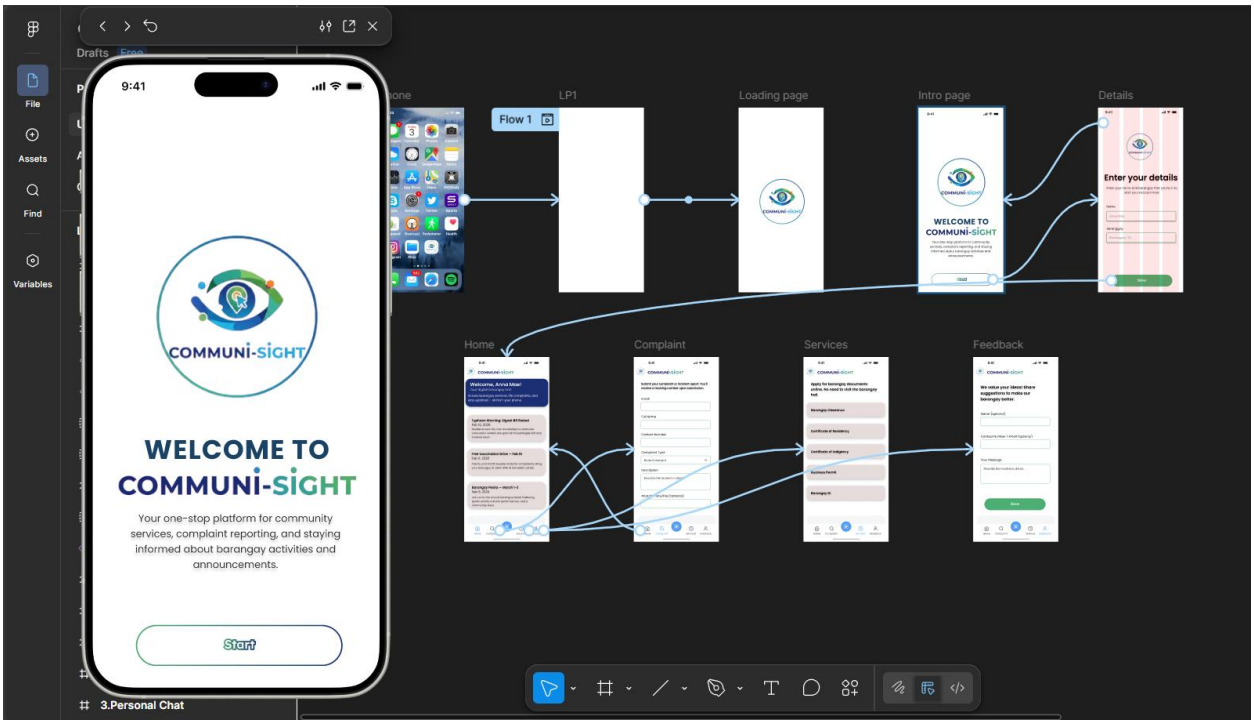


- **Prototyping:** Clickable prototypes using tools like Figma or Adobe XD.





- **High-Fidelity UI:** Final designs incorporating client branding (colors, typography)



5. Technical Approach

5.1 Tech Stack

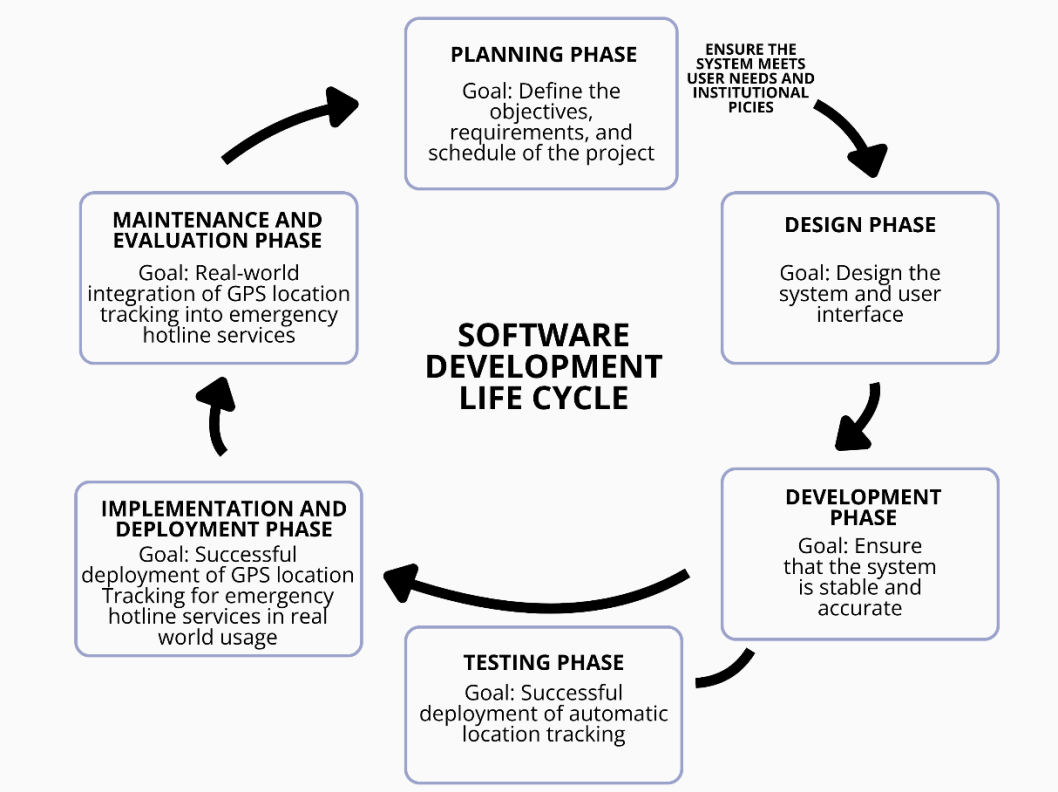
The front-end of the application may be developed using technologies such as React Native to enable efficient cross-platform development for both iOS and Android. The back-end may use technologies such as Node.js to support scalable and real-time data management. A database system such as PostgreSQL will store user data and reports securely. Cloud hosting services like AWS will be utilized to ensure reliability, performance, and data security.

Component	Technology Choice	Why we chose it
Front-End	React Native	Cost-effective cross-platform development for iOS and Android.
Back-End	Node.js	Handles real-time report updates and high user traffic efficiently.
Database	PostgreSQL	Reliable structured data for resident records and report history.
Cloud/Hosting	AWS	Industry-standard security and reliability for public data.

5.2 Security & Compliance

- **Data Encryption:** The app will implement data encryption using SSL/TLS to protect information during transmission.
- **Authentication:** Secure API authentication methods such as OAuth 2.0 will be used to ensure that only authorized users can access system resources.

6. Project Roadmap & Timeline



- **Phase 1: Discovery (Weeks 1-2):** Gathering specific barangay requirements.
- **Phase 2: UI/UX Design (Weeks 3-5):** Finalizing the app's look and resident flow.
- **Phase 3: Development (Weeks 6-12):** Building the reporting engine, API, and admin dashboard.
- **Phase 4: Testing (Weeks 13-14):** Rigorous QA and User Acceptance Testing with local residents.
- **Phase 5: Launch (Week 15):** App Store and Google Play Store submission.

7. Commercials (Investment)

7.1 Cost Breakdown (Cost of the Project)

Deliverable	Estimated Hours	Cost (USD)
UI/UX Design	80	₱187,168.00
Front-End Development	200	₱584,900.00
Back-End Development	150	₱438,675.00
QA & Testing	60	₱140,376.00
Total Project Cost		**₱1,351,119.00**

7.2 Payment Schedule

- **30% (₱405,335.70)** upon signing (Mobilization fee).
- **30% (₱405,335.70)** upon design approval.
- **20% (₱270,223.80)** upon Beta release.
- **20% (₱270,223.80)** upon final launch.

8. Maintenance & Support

Apps require updates. Define what happens after launch.

- **Warranty Period:** [e.g., 30 days] of free bug fixing post-launch.
- **Annual Maintenance Contract (AMC):** Optional monthly fee for OS updates, server monitoring, and minor tweaks.

9. Terms & Next Steps

- **Validity:** This proposal is valid until [Date].
- **Next Steps:** To proceed, please sign the attached agreement/contract and return it by [Date]. We will then schedule the kickoff meeting.